WRITE TIME once per new memory · a few small calls, about 4–5k input tokens, paid once

New memory

m_2

"Recommended a Seattle gym membership to Priya."

retrieve the 12 nearest rules

m_2

(a) Route LLM · 2 passes

candidate rules as option letters

A p_2 gym rule

B p_3 weekend-plan rule

C p_1 restaurant rule

D p_4 renewal rule

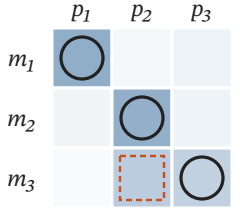
⋮ 8 more

q_m: next-letter probabilities
two option orders, averaged

q

(b) Assign no LLM

memories that name Priya



max Σ log q, ≤ 1 per rule
m_3: greedy p_2, joint p_3

p_2

(c) Ground LLM · 3 samples

condition of p_2, as OR of ANDs

city ∧ (exercises ∨ employer pays)

↓ each to one of 16 nearest items

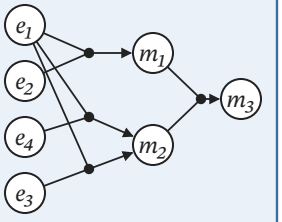
$J(m_2) = \{e_1, e_4\} \vee \{e_1, e_3\}$

s_1 ✓ s_2 ✓ s_3 ✓

sets in ≥ 2 of 3 samples kept;
no support → one retry, then reported

J

(d) Graph stored



+ source reliability
built incrementally

J (read only)

UPDATE TIME once per update · one LLM call of about 1.4k input tokens, flat in store size

Update

u_1

"I just finished settling into my new place in Austin."

source: user, in conversation

u_1

(e) Read LLM · 1 call

12 nearest facts

e_1 Priya lives in Seattle.

e_2 Priya is vegetarian.

e_4 Priya runs 3× a week.

⋮

target	e_1
relation	**supersedes**
source	**high**

relation from what is claimed

e_1, ρ

(f) Effect no LLM

status of the target e_1

older than e_1 — ignore

hedged / weaker — UNK

retracts — OUT

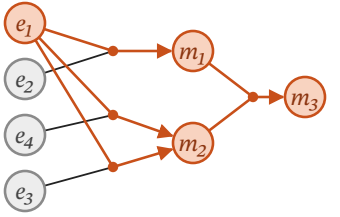
supersedes — STALE

first matching row applies

$\sigma(e_1) =$ STALE

σ

(g) Propagate no LLM · ms



$\sigma(m) = \max_{A \in J(m)} \min_{x \in A} \sigma(x)$

OUT < UNK < STALE < IN

σ

(h) Actions to caller

m_1	recompute
m_2	recompute
m_3	recompute
others	keep

if only u_2 arrived (weak):
e_3 uncertain, m_2 kept
via $\{e_1, e_4\}$

LLM call · deterministic · stored or returned · IN · keep · STALE · recompute · UNK · uncertain · OUT · withdraw